Scientific Reports

https://doi.org/10.1038/s41598-026-62286-9

Article in Press

The collective turing test: large language models can generate realistic multi-user discussions

Received: 27 October 2025

Accepted: 10 July 2026

Published online: 17 July 2026

Cite this article as: Bouleimen A., Marzo G., Kim T. *et al.* The collective turing test: large language models can generate realistic multi-user discussions. *Sci Rep* (2026). https://doi.org/10.1038/s41598-026-62286-9

Azza Bouleimen, Giordano Marzo, Taehee Kim, Nicolò Pagan, Hannah Metzler, Silvia Giordano, Anikó Hannák & David Garcia

We are providing an unedited version of this manuscript to give early access to its findings. Before final publication, the manuscript will undergo further editing. Please note there may be errors present which affect the content, and all legal disclaimers apply.

If this paper is publishing under a Transparent Peer Review model then Peer Review reports will publish with the final article.

ARTICLE IN PRESS

© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

ARTICLE IN PRESS

The Collective Turing Test: Large Language Models Can Generate Realistic Multi-User Discussions

Azza Bouleimen[1], Giordano De Marzo[2], Taehee Kim[2], Nicolò Pagan[1], Hannah Metzler[3], Silvia Giordano[4], Anikó Hannák[1], and David Garcia[2,3,*]

[1]University of Zurich, Department of Informatics, Zurich, 8050, Switzerland
[2]University of Konstanz, Department of Politics and Public Administration, Konstanz, 78464, Germany
[3]Complexity Science Hub, Vienna, 1030, Austria
[4]University of Applied Sciences and Arts of Southern Switzerland, Department of Innovative Technologies, Viganello, 6962, Switzerland
[*]david.garcia@uni-konstanz.de

ABSTRACT

Large Language Models (LLMs) offer new avenues to simulate online communities and social media. Potential applications range from testing the design of content recommendation algorithms to estimating the effects of content policies and interventions. However, the validity of using LLMs to simulate conversations between various users remains largely untested. We evaluated whether LLMs can convincingly mimic human group conversations on social media. We collected authentic human conversations from Reddit and generated artificial conversations on the same topic with two LLMs: Llama 3 70B and GPT-4o. When presented side-by-side to study participants, LLM-generated conversations were mistaken for human-created content 39% of the time. In particular, when evaluating conversations generated by Llama 3, participants correctly identified them as AI-generated only 56% of the time, barely better than random chance. Our study demonstrates that LLMs can generate social media conversations sufficiently realistic to deceive humans when reading them, highlighting both a promising potential for social simulation and a warning message about the potential misuse of LLMs to generate new inauthentic social media content.

ARTICLE IN PRESS

Introduction

Agent-Based Modeling (ABM) approaches have been used to identify principles that drive the emergence of collective behavior such as political polarization[1], segregation[2], collective emotions[3] cultural dynamics[4] and collective action[5]. While ABM can be used to illustrate generative sufficiency[6], i.e. the minimal conditions for a social phenomenon to emerge, they tend to oversimplify human behavior [7], leading to limited application in policies or interventions. To increase the practical applicability of ABM in policy-making or system design, the dynamics of individual behavior and interactions between agents need to be grounded on empirical data [8]. However, data scarcity and the variance of human behavior complicates the task of model calibration and limits the ability of ABM simulations to generate predictions.

Recent advances in generative AI, in particular Large Language Models (LLMs), promise to narrow this gap as they demonstrate high role-play capabilities[9–12]. LLMs can enable a new kind of support for the study and simulation of collective behavior[13,14] by complementing empirical data and allowing agents to communicate through real text or to decide agent actions based on LLM inference[15].

Social media are gaining significant traction as a prime application of social simulation. In light of the harm spreading on social media platforms[16,17] and their alarming real-world consequences[18–23], research on interventions and policies is needed to understand how to moderate the conversation and information flow on social media platforms. However, some intervention techniques[24] fail to mitigate online social issues[25,26], and can sometimes even cause further harm themselves[27–30].

Given the ethical concerns of conducting large-scale experimentation on social media[31], generative ABM, i.e. the usage of generative agents in simulations, presents itself as an ideal safe and controlled environment where researchers and policymakers can design and test different intervention strategies. Simulating social media through generative ABM has become even more pressing in recent years in light of the restrictions to public data access that social media platforms have introduced[32].

Generative ABM has been previously used to simulate social media platforms and the interest they spark is on the rise. Early studies show that LLMs can link forming online social networks similar to those created by humans[33]. More recently, they have been employed to test how different feed algorithms affect the level of toxicity and cross-partisan interactions[34–36]. Moreover, several different frameworks for simulating social networks have been developed[36–38]. For example, recent work by Ferraro et al.[36] provides a comprehensive framework for building specific personas as agents in a simulated social media environment. Despite this growing interest, there are still many open questions connected to the use of generative ABM. One of

ARTICLE IN PRESS

the most pressing is the need for novel validation approaches that show whether generative agents represent social media users, react in ways comparable to humans, and create overall group behavior that is akin to what humans do on social media. In other words, we still need to investigate the extent to which LLMs can mimic human behavior on social media.

Some studies have given initial insights into how conversations can be simulated with LLMs. Park et al.[39] introduced a prototyping technique to generate online forum discussions using GPT-3. For validation, they recruited crowdworkers to manually author comments on posts following authentic Reddit user comments. Both the crowdworker created conversations and the GPT-3 conversations were presented to a separate group of participants, who were asked to identify which conversation were AI-generated. The results showed that participants correctly identified AI-generated conversations 59% of the time, indicating that GPT-3 can be suitable for simulating social media-like conversations. We note, however, that the crowdworker-created conversations used for comparison in this experiment were not authentic Reddit conversations. This limits the external validity of the experiment, as the control condition of crowdworker conversations is likely not representative of true online interactions. To improve on this approach, we suggest in our paper an experiment based on genuine Reddit conversations, created by active Reddit users.

Pagan et al.[40] present a computational framework to systematically evaluate differences between LLM-generated social media and human replies. Their results reveal persistent divergences between human and LLM-generated replies, especially in affective and stylistic dimensions. However, they also showcase clear gains in realism from stylistic conditioning, context-aware prompting, and fine-tuning. While their approach is purely analytical, relying on BERT models and other Machine Learning (ML) systems to distinguish between LLM and human replies, our study employs an experimental design with human participants. This approach is complementary to Pagan et al.[40], as insights provided by humans are likely to be different from what ML models detect. Besides the studies mentioned above, the overall reliability of generative ABM to simulate online conversations among various individuals remains under-explored. In a recent survey, Larooij and Törnberg[13] find that validation remains poorly addressed in generative social simulations, and that even the most rigorous approaches fail to adequately demonstrate operational validity.

In this article, we present the results of a **Collective Turing Test**: Whether LLMs can generate the text of human-like conversations on social media. 75 years ago, Alan Turin described the Turing test to evaluate if a machine can generate a textual conversation that humans cannot distinguish from another human more than 30% of the time[41]. Here, we extend that concept to a multi-user discussion between machines with a human as an observer, asking how often would LLMs generate conversations that humans cannot distinguish from other human-generated conversations. We preregistered this study[42] as the first experiment that uses real Reddit conversations to test the capability of LLMs to mimic human conversations on social media. Specifically, we present human participants with a set of Reddit posts followed by two side-by-side conversations of identical length and structure. One conversation is an authentic human conversation as it took place on Reddit, while the other is artificially generated by LLMs. Participants are asked to identify which is the human conversation. We generated AI conversations using two LLMs, GPT-4o and Llama 3 70B, and presented the participants with conversations of varying length. Our first hypothesis is that the participants' success rate for conversations generated by Llama 3 70B will be lower than that of conversations generated with GPT-4o, i.e. that Llama 3 can generate more realistic conversations with the informality of Reddit users. Llama has been trained with internet-scale corpora that reportedly include high volumes of online forum-style dialogue, possibly aligning better with Reddit's tone and structure. Recently, it has been shown that Llama exhibits stylistic traits (such as tone, friendliness, and humor) that align well with human preferences in informal or social settings[43]. Second, we expect that longer conversations will be easier for participants to distinguish than shorter ones as longer threads provide more opportunities for participants to identify inconsistencies, unnatural flow, or semantic drift typical of AI-generated responses. However, this relationship may not necessarily be linear.

Results

A total of 251 participants took part in our study (see Methods for more details), generating 1741 annotations of pairs of authentic vs simulated discussions. We excluded all entries from 46 participants who failed our attention checks as well as the annotations corresponding to the attention checks. The remaining dataset contains 1205 annotations from 203 participants, satisfying the minimum sample size for 80% statistical power. We define *success rate* as the proportion of correctly identified human conversations by the participants. Overall, participants' success rate was 0.61. In other words, LLMs generated discussions that were mistaken to be human-made in 39% of the cases.

The impact of model and conversation length on the success rate

In this section we report findings about our preregistered hypotheses. The results of the multilevel logistic regression are presented in Fig. 1 and details about regression results are reported in Supplementary Table S1. There is a significant effect of the LLMs used to generate AI conversations on the participants' ability to correctly identify human conversations. More precisely, participants had a lower success rate in identifying human conversations when they were presented with simulations

ARTICLE IN PRESS

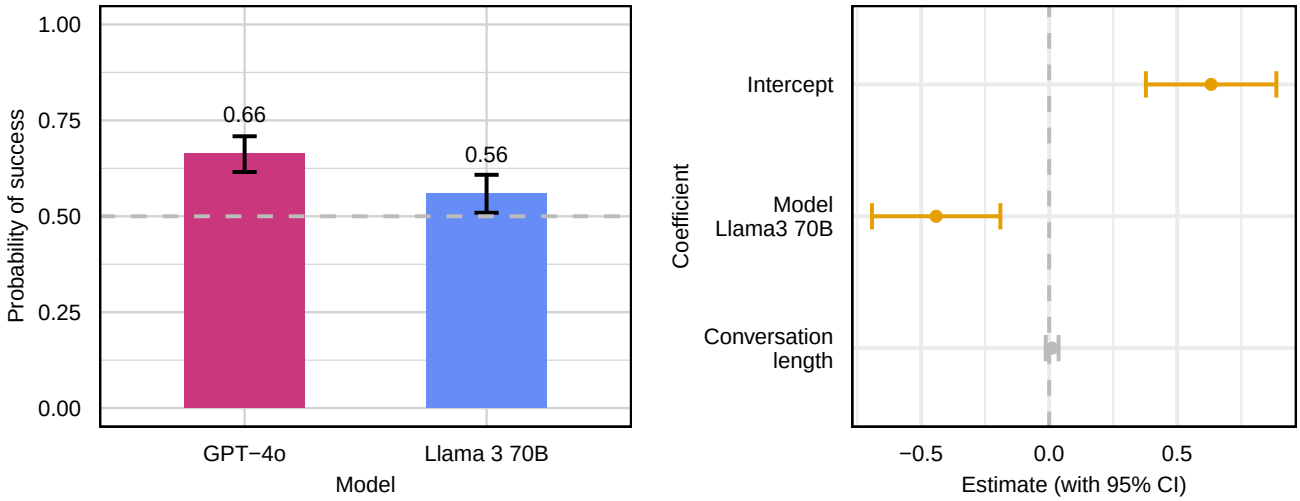


Figure 1. *Left:* Predicted probabilities of correctly identifying the human conversation by model. *Right:* Estimates of fixed effects in the logistic regression model with length as a numerical variable. 95% confidence intervals are shown and bars in orange are significant at the 95% level.

generated with Llama 3 70B than with GPT-4o (0.56 vs 0.66, $P < 0.001$). To account for potential heterogeneity across conversations, we additionally conducted a robustness check by including conversation identity as a random effect in the regression model; this yielded no substantive changes in the results (see Supplementary Table S5). Based on these observations, we conclude that Llama 3 70B is better at mimicking the text of human social media-like conversations than GPT-4o. In fact, Llama 3, as a locally deployable model, may generate more organic and less polished responses than GPT-4o, which is optimized for reliability and safety in commercial applications as a chatbot. Furthermore, Llama 3's training data likely includes social media conversations, given that it was developed by Meta. These characteristics may make Llama 3 outputs appear more human-like in informal online discussions.

We also hypothesized that longer conversations are easier to identify than shorter ones. Results show no significant effect of conversation length on participant success rates ($p - value = 0.36$) when conversation length is included in the regression as a numerical variable, thus we cannot reject the null hypothesis that longer conversations are easier to spot than shorter ones. To further investigate the role of conversation length, we conducted an additional analysis treating length as a categorical variable to test for nonlinear effects. The results of this second model are presented on Fig. 2 and details can be found on Supplementary Table S2.

When treating conversation length as a categorical variable, we find that length generally has no significant effect on success rates, except for length = 8, which is associated with a significantly higher success probability compared to length = 1. Fig. 2 (left for GPT-4o conversations and right for Llama 3 70B conversations), shows the predicted probabilities of correctly identifying the human conversation by model and conversation length. For both models, results shows that as conversation length increases, success rates increase up to a certain point (in this case, length = 8), beyond which success rates decline (please note that only length = 8 is significant). This preliminary observation aligns with the intuition that very short conversations are harder to distinguish, while longer conversations provide participants with more opportunities to detect characteristic patterns of AI-generated content. LLMs have been shown to exhibit lower variability in their generated text compared to humans[44], which may contribute to more uniform patterns that become detectable as conversation length increases. However, once conversation length reaches 16 posts, participant performance approaches random chance, suggesting that very long threads may make it harder to maintain focus. Given that only length = 8 coefficient is statistically significant, more investigations is needed in the direction of these observations.

Differences between human and AI conversations

As part of our preregistered exploratory analysis, we asked participants which cues helped them identify the AI-generated conversations after the annotation phase. Participants could freely provide their answers in a text box with few lines of space. Of the 203 participants who passed the attention test, 199 submitted a response to this question. To develop a codebook for classifying the responses, two researchers manually annotated an initial random subset of 50 answers. They identified three main categories: **Format**, **Style**, and **Content**. After discussion, the two researchers reached agreement on the scope of these

ARTICLE IN PRESS

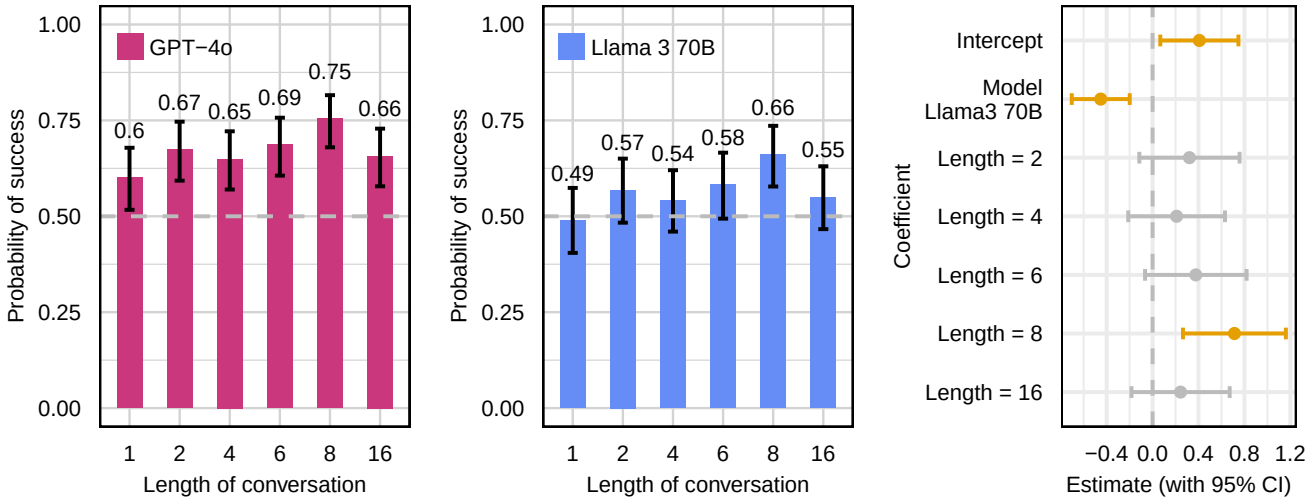


Figure 2. *Left:* Predicted probabilities of correctly identifying the human conversation by model = GPT-4o and length as a categorical variable. *Center:* Predicted probabilities of correctly identifying the human conversation by model = Llama 3 70B and length as a categorical variable. *Right:* Plot of the fixed effect of the logistic regression when length is a categorical variable. Reference is Length = 1 comment. Coefficients with the orange CI are statistically significant.

categories and their respective subcategories. Details are provided in Table 1. **Format** refers to visual cues in the text that can be noticed without reading it or knowing the conversational context. **Style** includes comments related to tone, voice, and stylistic choices. **Content** encompasses hints derived from the information itself, such as the level of explanation or argumentation provided.

We restricted the qualitative analysis to the subset of participants with a success score of ≥ 0.8 in identifying human conversations, resulting in 58 participants. Since these participants performed best in the task, their responses are more likely to convey reliable insights into the differences between AI and human conversations. Both researchers independently annotated this subset of answers according to the taxonomy described in Table 1. A single comment could refer to multiple categories and subcategories. The Cohen's Kappa scores for inter-annotator agreement at both category and subcategory levels are reported in the Supplementary Table S4 (Cohen's Kappa scores for the categories range between 0.66 and 0.75). After annotation, the two researchers met to discuss discrepancies and reached consensus on the final classification of the 58 comments. Table 2 shows the percentage distribution of categories and subcategories.

Textual **style** features served as the primary indicator participants used when distinguishing human from AI conversations, with 82.5% of responses mentioning at least one feature from this category. Within this category, most comments mentioned differences in *text formality* (49.1% of total comments), *typos and slang* (21.1%), followed by *politeness* and *language diversity* (each at 14%). This finding aligns with previous studies showing that language models, especially assistant-oriented ones, use considerably less toxic language than humans[40].

Content-related features ranked second, mentioned in 49.1% of comments. The most frequent subcategories in this category were *conformity* (19.3%), *explanations* (15.8%), and *authenticity* (12.3%). This latter observation suggests that, in our experiment, the models did not showcase authenticity through personal stories and anecdotes, in comparison to real social media conversations.

Format-related indicators ranked third, appearing in 17.5% of the participant comments. Most comments in this category mentioned response *length* as an identifying feature, found in 14% of all comments. Indeed, several participants stated things like "... the answers were longer...", "Too verbose..." and "Large amount of text..." about conversations they assumed to be AI-generated. This represents a particularly interesting observation given that LLMs were explicitly instructed to generate responses with approximately the same word count as their human counterparts. Considering that the second most frequent subcategory under **Content** was *explanations*, we hypothesize that LLMs remain overly verbose compared to humans, even when producing comments of similar length on average. They may convey less information than humans within the same amount of text.

ARTICLE IN PRESS

Format	Style	Content
• *emojis:* comments regarding the presence or absence of emojis in conversations. • *structure:* comments addressing structural aspects of sentences and text organization. • *length:* comments concerning the relative length of conversational responses.	• *tone:* comments noting the absence of sarcasm, irony, or humor in AI responses. • *text formality:* comments characterizing AI conversations as unnatural, overly polished, professional, or robotic. • *typos and slang:* comments observing that AI conversations contain fewer typographical errors, grammatical mistakes, or domain-specific slang compared to human conversations. • *affective responses:* comments highlighting the absence of profanity or other forms of emotional language in AI conversations. • *politeness:* comments describing AI text as excessively courteous or polite. • *language diversity:* comments identifying repetitive text patterns, linguistic monotony, or insufficient response variation.	• *objectivity:* comments characterizing AI responses as purely factual without subjective elements. • *explanations:* comments describing AI conversations as verbose or providing excessive detail and explanations. • *controversy:* comments noting AI's tendency to avoid controversial or clear-cut positions and maintain neutral stances. • *conformity:* comments identifying lack of personality and excessive agreement with users in AI responses. • *contextual misalignment:* comments indicating that AI replies were inappropriate for the given context. • *general positivity:* comments observing that AI conversations exhibit greater positivity and enthusiasm compared to human conversations. • *authenticity:* comments identifying generic response patterns and absence of genuine personal opinions in AI conversations.

Table 1. Categories and subcategories of the comments of participants on how they spotted the AI conversations.

Discussion

In this study, we conducted the first experiment investigating the capacity of large language models (LLMs) to mimic the collective behavior of humans on social media platforms using real Reddit data. To generate AI conversations, we used a minimal prompting strategy, asking the models to behave like Reddit users without assigning any specific personas (the exact prompt is provided in the Supplementary Information). Despite this simplicity, participants failed to identify the human conversation in 39% of cases. This exceeds the 30% misclassification threshold traditionally associated with passing the Turing Test, suggesting that even basic prompting can yield conversations that convincingly imitate human interactions.

Notably, conversations generated by **Llama 3 70B** were significantly harder to distinguish from human ones than those produced by **GPT-4o**, pointing to meaningful differences in how specific models simulate human-like discourse. To assess a potential limitation related to participants' familiarity with Reddit, we conducted an additional analysis (reported in the Supplementary Information) examining whether familiarity predicts detection performance. This analysis reveals that participants familiar with Reddit achieved higher success rates than those who were not (difference ≈ 0.1; see Supplementary Table S6), and the effect is statistically significant. This suggests that familiarity with platform-specific norms, styles, and language may improve participants' ability to detect AI-generated content. This highlights a limitation of our study, as variation in participants' familiarity with Reddit may influence detection performance. Nevertheless, these findings remain *promising* for researchers seeking to use LLMs in ABMs to simulate social systems more realistically, as they suggest that agents powered by LLMs can convincingly emulate human dialogue. However, the results are also *concerning*, as they highlight the risk of LLM-generated content blending into genuine online discourse undetected. This could be exploited to create artificial consensus or engineer social polarization through coordinated "AI swarms", as previously suggested in the literature[45].

Surprisingly, **conversation length** did not significantly impact participants' ability to distinguish between AI and human threads. However, we observed an *inverse-U relationship*, where detection accuracy initially improved with conversation length peaking at 8 comments before declining. This non-linear pattern suggests that longer conversations, up to a certain length, may introduce enough natural-sounding variance to mask the LLM origin, or may lead to participant fatigue. Both hypotheses warrant further investigation.

ARTICLE IN PRESS

Category	Category Percentage	Subcategory	Subcategory Percentage
Format	17.5%	emojis	1.8%
		structure	7.0%
		length	14.0%
Style	82.5%	tone	12.3%
		text formality	49.1%
		typos and slang	21.1%
		affective responses	12.3%
		politeness	14.0%
		language diversity	14.0%
Content	49.1%	objectivity	8.8%
		explanations	15.8%
		controversy	3.5%
		conformity	19.3%
		contextual misalignment	1.8%
		general positivity	1.8%
		authenticity	12.3%

Table 2. Distribution of the categories and subcategories of hints mentioned in the participant's answers on what made them identify the AI-conversations.

Our **qualitative analysis** of participant responses revealed several recurring cues used to assess authorship. Many noted that AI-generated conversations were overly *polite*, *formal*, and devoid of slang or profanity, features that are commonplace in authentic Reddit discourse. This suggests a limitation of current LLMs in simulating toxic or emotionally charged behavior, which is crucial for studying online conflict or misinformation. Participants also observed a lack of *authenticity*, emotional nuance, and personal storytelling in AI responses, which reduced the realism of emotional or persuasive discourse. Additionally, a frequent comment was the tendency for AI users to *over agree* with one another, which could compromise the validity of simulations focused on disagreement or polarization. Taken together, these insights provide a promising foundation for future work on prompt design aimed at improving the realism of generated conversations.

These findings highlight both the potential and the limitations of current LLMs in replicating human behavior in online settings, particularly in generative ABM. While our study shows that LLMs can reproduce certain surface-level textual features of human interaction, it also reveals that LLM-generated content tends to lack key aspects of social media behavior, such as conflict, emotional intensity, and authentic self-expression. These limitations are particularly relevant for ABM applications that aim to simulate polarization, disagreement, or the spread of misinformation, where such dynamics play a central role. Improving realism in these contexts will likely require careful prompt engineering, including persona-driven strategies, and model selection. Our study also does not address other important components of ABM, such as personas, behavioral triggers, or network structures, in the context of LLM-based simulations. Evaluating these aspects remains an important direction for future research.

Conclusion

Our findings offer three key takeaways:

1. **LLMs can convincingly simulate human discourse on social media.** Across models and conversation lengths, human evaluators mistook LLM-generated Reddit conversations for human-authored ones in 39% of cases, surpassing the Turing Test threshold. This highlights the potential of LLMs for generative agent-based modeling (ABM).

2. **Model selection significantly affects performance.** Llama 3 70B outperformed GPT-4o in producing more human-like conversations. Based on these results, we recommend Llama 3 70B for use in simulations aiming to replicate naturalistic human interaction.

3. **Limitations persist in simulating emotional and polarizing dynamics.** Although LLMs often pass as human, they struggle to replicate emotionally charged discourse, personal storytelling, or polarization, factors that are central to many social media phenomena.

ARTICLE IN PRESS

Future work should extend this "Collective Turing Test" approach to include additional LLMs, such as proprietary models like Claude and open-weight models like Mistral. In particular, it will be valuable to investigate whether the effect holds with recent reasoning models. While chain-of-thought reasoning models may improve coherence and adherence to instructions, it remains unclear whether these capabilities translate into more human-like behavior in social contexts, or whether they address the limitations identified in our study, particularly those related to emotional and polarizing dynamics, which are not primarily driven by structured reasoning. Such evaluations could serve as benchmarks for selecting models in social simulations. Moreover, the study can be replicated in other languages, leveraging Reddit's multilingual data. Another promising direction involves using multiple persona-driven LLM agents in conversations to assess whether this setup improves authenticity, fosters personal storytelling, or generates more emotionally intense or controversial discourse.

In summary, LLMs exhibit substantial capabilities in mimicking online human behavior, but important limitations remain. Researchers must apply these tools critically, especially when simulating sensitive phenomena such as polarization, misinformation, or emotional contagion. We hope this work supports the growing community of computational social scientists by offering empirical benchmarks, practical model comparisons, and methodological guidance for conducting LLM-based social simulations.

Methods

To build and conduct our experiment, we first collected real human data from Reddit and then generated parallel AI conversations. After that, we built the experiment environment and ran the experiment with online participants, following our preregistration.

Collecting and generating data for stimuli

We sampled Reddit data from Pushshift dataset, which contains an exhaustive compilation of public Reddit discussions up to December 2023[46]. We selected conversations from December 2023 to avoid including data potentially present in the training sets of GPT-4o or Llama 3 70B, as their training cutoffs are Oct. 1st 2023. and Dec. 1st 2023, respectively[47,48]. From the Reddit posts we collected, we selected 16 posts spanning 8 different topics: food, leisure, opinions, relationships, technology, movies, health, and sports. The exhaustive list of selected Reddit posts is presented in Supplementary Table S3. For each post, we scraped the first 20 comments as they appear to a logged-in user. Reddit's comment ranking algorithm is user-independent and relies on comment-specific metrics such as upvotes and downvotes[49]. Consequently, all logged-in users see the comments in the same order, allowing us to replicate what typical Reddit user experience. We anonymized the data by replacing usernames and profile pictures with fictional alternatives. From the scraped comments, we extracted their conversational structure: which user replied to which other user under which thread of comment, and word counts for each comment. In other words, we kept the tree-structure of the conversation fixed, while we replaced each human comment with an AI generated one. In doing so, we provided the LLMs with the length (in number of words) of the original comment to preserve the length of comments. This way, we only focus on the conversation content, while leaving structure and length fixed.

To generate the artificial conversations, we prompted both models with the text reported in Supplementary Information, "Model prompt". We did one round of refinement of the prompt following feedback from a pilot study and left the prompt unchanged after we preregistered our experiment. For this study, we did not attempt to create distinct personas when prompting the LLMs. All LLM instances shared the same standard model configurations with additional instruction on behaving as Reddit users. No conversational examples were previously provided to the models. For each comment to generate, we created a new LLM instance and provided it with the prompt, the post text, and all comments generated so far by previous LLM instances. Specifically, we wrote:

```
Write a reply with about {word_count} words to comment ID {in_reply_to_id}.
```

The model was then tasked with generating text of approximately the same length as the corresponding human comment (`word_count`), either in reply to the corresponding user or as a direct comment on the post (represented by `in_reply_to_id`). We generated conversations using GPT-4o and Llama 3 70B in early Dec. 2024. For each model, we used three different temperatures: 0.2, 0.7, and 1.2. The same conversation could then be shown to participants at 6 different lengths, consisting of the first 1, 2, 4, 6, 8, or 16 comments from the generated conversations. Following this procedure we obtained 574 different stimuli combining posts with AI versus human conversations for presentation to participants (2 LLM models × 3 temperatures × 6 lengths × 16 posts). These conversations were generated once before our experiment preregistration and were not regenerated afterwards.

Experimental setting

We used the same HTML layout as Reddit's webpage to generate images of both AI and human conversations. For the annotation task, we relied on Potato[50], a Python library for building annotation systems, which we adapted to the needs of our experiment. Each participant first viewed an information page about the experiment (see Supplementary Information, "Information page").

ARTICLE IN PRESS

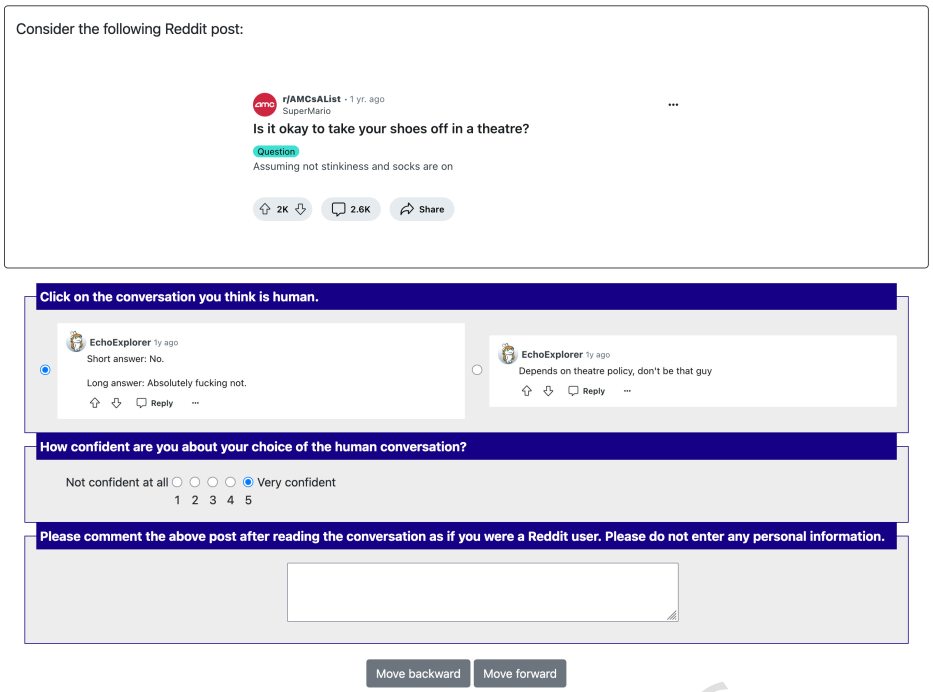


Figure 3. Example of annotations shown to participants.

This page clearly stated that participants will see two conversations in every task, and that one is AI-generated while the other is human. After that, the experiment started with a pre-screening survey where participants provided informed consent and answered two basic attention-check questions (see Supplementary Fig. S1). Participants who failed either question were not allowed to proceed and were directed to an exit page, while those who succeeded advanced to the annotation phase. In this phase, each participant completed 6 annotations and 1 additional attention test (an example is provided in Supplementary Fig. S2). To prevent redundancy, we extended the Potato library with a custom sampling algorithm ensuring that no participant saw the same Reddit post more than once, even if presented with different conversation lengths, LLM models, or temperature settings.

To avoid acquiescence bias[51], every annotation task included two conversations, side-by-side, one human and one AI generated. An example of the interface presented to participants is shown in Fig. 3. The two conversations have the same number of comments from the same number of users involved, having the same usernames and profile pictures. The only thing that differs between the human and the AI conversations is the text of the comments, which was made to be approximately the same number of words. We asked the participant to select which conversation was human-made. When doing, so the other conversation was greyed out. Participants were then able to report their level of confidence in their answer. Finally, we asked them to comment on the post as if they were authentic Reddit users taking part in the conversation. We hypothesized that participants were not always carefully reading the comments. Therefore, we added this last task in an attempt to get them more involved in the conversation as a real Reddit user would be. After completing the annotation phase, participant were redirected to a post-survey questionnaire, in which we asked participants to write down how they distinguished the human from the AI conversation. This was done in order to provide a qualitative assessment on how to better prompt the models to mimic human conversations on social media. We also ask participants questions about their familiarity with Reddit and with AI assistants. For the scope of this paper, we do not report the results of the former questions. The design of this experiment was approved by the Ethics Committee of the University of Konstanz (IRB statement 10/2025). All methods and experiments were performed in accordance with the Declaration of Helsinki, the specific regulations of the University of Konstanz for ethical experimentation, and the relevant national and international law and regulations.

We preregistered our study with two hypotheses and four exploratory analyses: the potential interaction effect between the LLM model and the length of the conversation[42]. Note that two deviations from the preregistration happened. First, there was a typo in the text of the first hypothesis. Instead of stating that our hypothesis was that Llama 3 70B would have a *lower* success rate, the preregistration text mistakenly stated *higher* success rate. However, the accompanying explanation of the hypothesis in the preregistration clearly supports the expectation of a lower success rate. This typo was due to the fact that, in an initial version of the preregistration we were reasoning from the deceiving rate perspective and not the success rate. Second, due to a

ARTICLE IN PRESS

technical issue, the two conversations from the topic *movies* were not included in the stimulus set from which participants' annotations were sampled. As a result, contrary to what was preregistered, the total number of stimuli effectively presented to participants was 504 (2 LLM models × 3 temperatures × 6 lengths × 14 posts) rather than 576. The power analysis indicated that a sample of 200 participants would be required to achieve 80% power. Accounting for the 20% failure rate on the attention test observed in the pilot, we recruited 250 English-fluent participants from Prolific. We compensated them with 9£/hour when we conducted the study in May 2025.

Statistical model

We fit a multilevel logistic model of the probability of participants identifying the AI-generated conversation correctly. Since participants respond repeatedly to up to 6 annotations in our experiment, the data is nested within participants. To account for this nested structure, we employ a multilevel model with varying intercepts for participants. The independent variables are:

- Length: A continuous variable with values 1, 2, 4, 6, 8, and 16, representing the number of comments in the conversation presented to participants. For example, a value of 2 indicates that respondents were shown two comments in the conversation related to the Reddit post.
- Model: A categorical binary variable indicating the AI model used (0 = GPT-4o vs. 1 = Llama 3 70B)

The dependent variable y is binary: correct guess (1: correctly spotting the human conversation) or incorrect guess (0: failing to spot the human conversation). The fitted the model using the R package `lme4` (`lme4`: Linear Mixed-Effects Models using 'Eigen' and S4) using the formula $y \sim model + length + (1|participant)$.

To further investigate the role of conversation length, we conducted an additional analysis treating length as a categorical variable to test for nonlinear effects, following the formula $y \sim model + factor(length) + (1|participant)$. Where $factor(length)$ represents the length of the conversation as a categorical variable.

References

1. Schweighofer, S. & Garcia, D. Raising the spectrum of polarization: Generating issue alignment with a weighted balance opinion dynamics model. *J. Artif. Soc. & Soc. Simul.* **27** (2024).
2. Schelling, T. C. Dynamic models of segregation. *J. mathematical sociology* **1**, 143–186 (1971).
3. Schweitzer, F. & Garcia, D. An agent-based model of collective emotions in online communities. *The Eur. Phys. J. B* **77**, 533–545 (2010).
4. Axelrod, R. The dissemination of culture: A model with local convergence and global polarization. *J. conflict resolution* **41**, 203–226 (1997).
5. Granovetter, M. Threshold models of collective behavior. *Am. journal sociology* **83**, 1420–1443 (1978).
6. Epstein, J. M. Generative social science: Studies in agent-based computational modeling. In *Generative Social Science* (Princeton University Press, 2012).
7. Flache, A. *et al.* Models of social influence: Towards the next frontiers. *Jasss-The journal artificial societies social simulation* **20**, 2 (2017).
8. Windrum, P., Fagiolo, G. & Moneta, A. Empirical validation of agent-based models: Alternatives and prospects. *J. Artif. Soc. Soc. Simul.* **10**, 8 (2007).
9. Grossmann, I. *et al.* Ai and the transformation of social science research. *Science* **380**, 1108–1109 (2023).
10. Hämäläinen, P., Tavast, M. & Kunnari, A. Evaluating large language models in generating synthetic hci research data: a case study. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 1–19 (2023).
11. Mei, Q., Xie, Y., Yuan, W. & Jackson, M. O. A turing test: Are ai chatbots behaviorally similar to humans? *Available at SSRN* (2023).
12. Bubeck, S. *et al.* Sparks of artificial general intelligence: Early experiments with gpt-4. *arXiv preprint arXiv:2303.12712* (2023).
13. Larooij, M. & Törnberg, P. Do large language models solve the problems of agent-based modeling? a critical review of generative social simulations. *arXiv preprint arXiv:2504.03274* (2025).
14. De Marzo, G., Castellano, C. & Garcia, D. Ai agents can coordinate beyond human scale. *arXiv preprint arXiv:2409.02822* (2024).
15. Xi, Z. *et al.* The rise and potential of large language model based agents: A survey. *Sci. China Inf. Sci.* **68**, 121101 (2025).

ARTICLE IN PRESS

16. Woolley, S. & Howard, P. *Computational propaganda worldwide: Executive summary* (Computational Propaganda Project, 2017).

17. Baumgartner, J. C. & Towner, T. L. *The Internet and the 2020 Campaign* (Rowman & Littlefield, 2021).

18. Budak, C. What happened? the spread of fake news publisher content during the 2016 us presidential election. In *The World Wide Web Conference*, 139–150 (2019).

19. Guess, A., Nyhan, B. & Reifler, J. Selective exposure to misinformation: Evidence from the consumption of fake news during the 2016 us presidential campaign. *Eur. Res. Counc.* **9**, 4 (2018).

20. Ferrara, E. Disinformation and social bot operations in the run up to the 2017 french presidential election. *arXiv preprint arXiv:1707.00086* (2017).

21. Yerlikaya, T. & Aslan, S. T. Social media and fake news in the post-truth era. *Insight Turk.* **22**, 177–196 (2020).

22. Islam, M. S. *et al.* Covid-19–related infodemic and its impact on public health: A global social media analysis. *The Am. journal tropical medicine hygiene* **103**, 1621 (2020).

23. Pierri, F. *et al.* Online misinformation is linked to early covid-19 vaccination hesitancy and refusal. *Sci. reports* **12**, 5966 (2022).

24. Smith, F., Simchon, A., Holford, D. & Lewandowsky, S. Inoculation reduces social media engagement with affectively polarized content in the uk and us. *Commun. Psychol.* **3**, 11 (2025).

25. Chandrasekharan, E., Jhaver, S., Bruckman, A. & Gilbert, E. Quarantined! examining the effects of a community-wide moderation intervention on reddit. *ACM Transactions on Comput. Interact. (TOCHI)* **29**, 1–26 (2022).

26. Trujillo, A. & Cresci, S. Make reddit great again: assessing community effects of moderation interventions on r/the_donald. *Proc. ACM on Human-computer Interact.* **6**, 1–28 (2022).

27. Bail, C. A. *et al.* Exposure to opposing views on social media can increase political polarization. *Proc. Natl. Acad. Sci.* **115**, 9216–9221 (2018).

28. Dias, N., Pennycook, G. & Rand, D. G. Emphasizing publishers does not effectively reduce susceptibility to misinformation on social media. *HKS Misinformation Rev.* (2020).

29. Zollo, F. *et al.* Debunking in a world of tribes. *PloS one* **12**, e0181821 (2017).

30. Horta Ribeiro, M. *et al.* Do platform migrations compromise content moderation? evidence from r/the_donald and r/incels. *Proc. ACM on Human-Computer Interact.* **5**, 1–24 (2021).

31. Kramer, A. D., Guillory, J. E. & Hancock, J. T. Experimental evidence of massive-scale emotional contagion through social networks. *Proc. Natl. Acad. Sci.* **111**, 8788–8790 (2014).

32. Chan, X., Wang, X., Yu, D., Mi, H. & Yu, D. Scaling synthetic data creation with 1,000,000,000 personas. *arXiv preprint arXiv:2406.20094* (2024).

33. De Marzo, G., Pietronero, L. & Garcia, D. Emergence of scale-free networks in social interactions among large language models. *arXiv preprint arXiv:2312.06619* (2023).

34. Törnberg, P., Valeeva, D., Uitermark, J. & Bail, C. Simulating social media using large language models to evaluate alternative news feed algorithms. *arXiv preprint arXiv:2310.05984* (2023).

35. Gao, C. *et al.* S ^{3}: Social-network simulation system with large language model-empowered agents. *arXiv preprint arXiv:2307.14984* (2023).

36. Ferraro, A. *et al.* Agent-based modelling meets generative ai in social network simulations. In Aiello, L. M., Chakraborty, T. & Gaito, S. (eds.) *Social Networks Analysis and Mining*, 155–170 (Springer Nature Switzerland, Cham, 2025).

37. Yang, Z. *et al.* Oasis: Open agent social interaction simulations with one million agents. *arXiv preprint arXiv:2411.11581* (2024).

38. Rossetti, G. *et al.* Y social: an llm-powered social media digital twin. *arXiv preprint arXiv:2408.00818* (2024).

39. Park, J. S. *et al.* Social simulacra: Creating populated prototypes for social computing systems. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, 1–18 (2022).

40. Pagan, N., Törnberg, P., Bail, C., Hannak, A. & Barrie, C. Can llms imitate social media dialogue? techniques for calibration and bert-based turing-test. In *First Workshop on Social Simulation with LLMs* (2025).

41. Turing, A. Computing machinery and intelligence. *Mind* **LIX**, 433–460 (1950).

ARTICLE IN PRESS

ARTICLE IN PRESS

42. Bouleimen, A. The collective turing test: Investigating the similarity between ai-generated and human online conversations. https://osf.io/uedq4/files/osfstorage (2025). Open Science Foundation. Accessed 15/08/2025.

43. Dunlap, L., Mandal, K., Darrell, T., Steinhardt, J. & Gonzalez, J. E. Vibecheck: Discover and quantify qualitative differences in large language models. *arXiv preprint arXiv:2410.12851* (2024).

44. Zanotto, S. E. & Aroyehun, S. Human variability vs. machine consistency: A linguistic analysis of texts generated by humans and large language models. *arXiv preprint arXiv:2412.03025* (2024).

45. Schroeder, D. T. *et al.* How malicious ai swarms can threaten democracy. *arXiv preprint arXiv:2506.06299* (2025).

46. Baumgartner, J., Zannettou, S., Keegan, B., Squire, M. & Blackburn, J. The pushshift reddit dataset. In *Proceedings of the international AAAI conference on web and social media*, vol. 14, 830–839 (2020).

47. OpenAI. Model release notes. https://help.openai.com/en/articles/9624314-model-release-notes (2025). OpenAI. Accessed 15/08/2025.

48. Meta-Llama. Model details. https://huggingface.co/meta-llama/Meta-Llama-3-70B (2025). Hugging Face. Accessed 15/08/2025.

49. Salihefendic, A. How reddit ranking algorithms work. https://medium.com/hacking-and-gonzo/how-reddit-ranking-algorithms-work-ef111e33d0d9 (2015). Medium. Accessed 15/08/2025.

50. Potato. Potato, the portable text annotation tool for everyone. https://potato-annotation-tutorial.readthedocs.io/en/latest/. Potato Annotation. Accessed 15/08/2025.

51. Billiet, J. B. & McClendon, M. J. Modeling acquiescence in measurement models for two balanced sets of items. *Struct. equation modeling* **7**, 608–628 (2000).

Acknowledgments

We would like to acknowledge the contributions of Peer Saleth for assisting in setting up the experiment server, as well as Salima Jaoua and Christopher Mayer for the constructive discussions on statistical analysis.

Funding statement

This research was funded by Ministry of Science, Research and the Arts of Baden-Württemberg under project 33966824. A.B. was partially funded by the Swiss National Science Foundation (SNSF) via the SINERGIA project CARISMA (grant CRSII5_209250), https://carisma-project.org. T.K. was funded by the European Union's Horizon Europe research and innovation programme under grant agreement No. 101177574 (WHAT-IF). Views and opinions expressed are those of the author(s) only and do not necessarily reflect those of the European Union. Neither the European Union nor the granting authority can be held responsible for them.

Author contributions statement

A.B. developed the experiment platform, collected data, conducted statistical analysis, conducted qualitative analysis and drafted the manuscript. G.D.M. produced experiment stimuli with AI simulations. T.K. contributed to research design and developed the statistical analysis framework. N.P. conducted qualitative analysis. H.M. contributed to research design. D.G. designed research and statistical analysis and supervised the project. All authors reviewed the manuscript.

Competing interests

The author(s) declare no competing interests.

Data availability statement

The code and anonymized data of our study can be publicly accessed at https://github.com/azza-bouleimen/turing-test-ai-conversations